

The Gauge Is the Obstacle:

Task Identity, Cross-Family Transfer, and Merge-Conflict Prediction from Adapter Weights Alone

(Workshop condensation)

Timothy Paul Bielec*
TASUMER MAF

Araminta K. Carlson
TASUMER MAF

August 2026

Abstract

Raw flattened weights are the unexamined default of adapter analysis; we measure what the default costs. Against a pre-registered bank of 480 LoRA adapters—2 model families $\times$ 6 tasks $\times$ 40 seeds—raw weights fail a locked task-identifiability test (leave-one-out accuracy 0.0792 / 0.1375, chance 0.1667), while $GL(r)$ -gauge-canonical and vocabulary-signature representations identify all six tasks at 1.0000 (permutation $p = 0.000999$). Our pre-registered prediction that task structure would *not* transfer across families was refuted by its own control: raw transfer-at-chance was a family-scale covariate-shift artifact (family-identity probe 1.0000 on raw features), and shift-controlled transfer runs 0.7375–0.7833 (binomial $p \leq 1.20 \times 10^{-84}$). One mechanism covers both: nuisance variation—the $GL(r)$ factorization gauge within a family, the scale gauge between families—dominates raw coordinates; task identity is gauge-obscured, not absent. Companion results: every bridge swap preserving the identity backbone costs under +0.001 nats, while destroying it costs +2.8086 / +3.8365; weight-only features predict midpoint-merge conflict at group-aware AUC 0.995 / 0.962; a pilot correlation of $r = 0.888$ shrinks honestly to 0.300. Hypotheses were locked before the bank existed, each analysis ran exactly once, every headline was independently re-derived, and two of five outcomes went against us.

1 The result, in ninety seconds

A trained LoRA adapter is a small stack of low-rank matrices, and adapters now accumulate by the thousands—shared on hubs, merged into products, composed by people who never saw the training run. Given only the adapter tensors—no training data, no logs, no access to the training process—what can be read out? The obstacle is not information but parameterization: a LoRA update factorizes as BA , and for any invertible $r \times r$ matrix M the reparameterization $(B, A) \rightarrow (BM, M^{-1}A)$ leaves the effective update unchanged—the $GL(r)$ gauge. Raw flattened weights are the unexamined default of adapter analysis, and this paper measures what the default costs.

The answer, measured once against a pre-registered 480-adapter bank: the default costs legibility itself. Raw adapter weights fail a locked task-identifiability test at leave-one-out accuracy 0.0792 (Qwen) / 0.1375 (Llama) against chance 0.1667—while the same adapters, canonicalized to remove the $GL(r)$ gauge, identify all six training tasks at 1.0000 in both families (permutation $p = 0.000999$), at a small fraction of the raw dimensionality. One level up, the identical story: our own pre-registered prediction that task structure would *not* transfer across model families was refuted by its own triviality control—raw transfer-at-chance was a covariate-shift artifact of family scale, and once an unsupervised per-family standardization removes it, task structure transfers between a 1.5B and a 1B model of different lineages at 0.7375–0.7833. Task identity is gauge-obscured, not absent. The canonicalization, not classifier capacity, is the binding constraint—scoped to coarse label granularity within a controlled family (Section 5).

*Corresponding author. timothy@promptcrafted.com. Affiliations are placeholders pending the venue author block.

Why a reader deciding quickly should care: (i) if you triage finished adapters post hoc—provenance, transfer, mergeability—canonicalize before you conclude anything; a linear probe on gauge-fixed features suffices where the raw default fails; (ii) the same weight-only regime supports a supervised merge-conflict predictor at group-aware AUC 0.995 / 0.962 with a CI-separated margin over raw distance; and (iii) the numbers arrive with unusual evidentiary standing—hypotheses locked before the bank existed, every open choice pinned in dated rulings, a completeness interlock that held, each analysis run exactly once, every headline independently re-derived by a separate Director instance, and two of the five pre-registered outcomes decided *against* the authors (Section 4).

2 The measurement

2.1 The instrument

The artifact is a bank of 480 LoRA adapters: 2 model families $\times$ 6 tasks $\times$ 40 seeds (Qwen2.5-1.5B-Instruct and Llama-3.2-1B-Instruct; tasks alpaca, code, math, xsum, squad, agnews). Every run trains a rank-24 adapter on the q/k/v/o projections with a 6-channel identity-initialized bridge per wrapped module (effective update $\Delta W = B(M \otimes I_4)A$; at identity initialization the bridge is exactly absent and the update is standard LoRA): AdamW at peak LR 0.0002, weight decay 0.01, clip 1.0, linear warmup (100 steps) then cosine decay to a 0.1 floor; 2000 optimizer steps at effective batch 16 (4×4) = 32,000 sequences of length 512; per-task pools capped at 40,000 examples and a fixed 500-example validation split per task (seed 777); bfloat16 base, float32 adapters. Adapter module counts are discovered from each adapter rather than hardcoded: 112 modules per run for Qwen, 64 for Llama. The campaign ran on a single local GPU from July 3 21:52Z to July 20 16:21Z—roughly 17 days including one restart—and the manifest closed at 480/480 COMPLETE on 2026-07-20T16:21:19Z; the outage ledger contains exactly four failures, all HF Hub 504 errors on July 16 (run indices 329–332), all retried to COMPLETE. Chance accuracy for the 6-task classification problems throughout is 0.1667.

2.2 H1: task identity is legible only after canonicalization

H1 asks whether an adapter’s weights identify the task it was trained on, within a family. The classifier is deliberately weak—a leave-one-out linear SVM ($C = 1.0$) over 240 adapters per family, with a 1,000-shuffle permutation null—so that the representation, not model capacity, carries the result. The pre-registered lock is two-sided: LOO accuracy above $1.5 \times$ chance (0.2500) **and** permutation $p < 0.01$. Four representations were computed per family: raw flattened weights; the $GL(r)$ -gauge-canonical representation (QR→SVD lineage, bridge absorbed); and two pre-registered vocab-signature variants (output-referenced through the base model’s unembedding), with and without k/v modules.

Table 1: Within-family task identifiability by representation. LOO accuracy, linear SVM, 1,000-shuffle permutation null. Lock: acc $> 1.5 \times$ chance (0.2500) AND perm $p < 0.01$. Chance = 0.1667.

family	representation	dim	LOO acc	perm p	H1 lock
qwen2.5-1.5b	raw	6,541,248	0.0792	0.000999	FAIL
qwen2.5-1.5b	canonical	88,704	1.0000	0.000999	PASS
qwen2.5-1.5b	vocab.signature	15,232	1.0000	0.000999	PASS
qwen2.5-1.5b	vocab.sig_kv_exclude	7,616	1.0000	0.000999	PASS
llama3.2-1b	raw	5,114,112	0.1375	0.000999	FAIL
llama3.2-1b	canonical	50,688	1.0000	0.000999	PASS
llama3.2-1b	vocab.signature	8,704	1.0000	0.000999	PASS
llama3.2-1b	vocab.sig_kv_exclude	4,352	1.0000	0.000999	PASS

The pattern is stark and consistent (Table 1). Raw adapter weights—the highest-dimensional representation—fail the lock at or below chance; the canonical representation identifies all six tasks perfectly, as do both vocab-signature variants, so the result is not specific to one canonicalization route. Raw’s tiny permutation p

alongside sub-threshold accuracy is not a contradiction: under LOO, label permutation drives accuracy *below* chance (raw nulls center at 0.012 Qwen / 0.057 Llama, maxima 0.054 / 0.121), so both raw arms exceed their null maxima and take the floor $p = 1/1001$ while carrying only a weak recoverable signal (per-class recall 0.45 on code for Qwen with every other class ≤ 0.025 ; 0.425 on alpaca plus 0.40 on code for Llama with every other class at 0). This is exactly the case the two-sided lock was built to reject—statistically distinguishable from noise, not usefully legible. A pre-registered variance-heterogeneity guard (ratios 1.00–1.46 against a trigger of 3.7) was not triggered: the canonical ceiling is not a variance artifact. The Director re-derived all eight LOO accuracies from the confusion matrices and re-ran the canonical LOO-SVM from scratch on the feature matrices, obtaining 1.0000 for both families.

2.3 H2: the pre-registered prediction, refuted by its own control

H2 was our boldest pre-registered prediction, and it was wrong. We predicted that task structure would *not* transfer across model families. The verdict is **NOT supported**—and the refutation was produced by our own triviality control, added in round-1 review before any data existed.

Two dimension-agnostic representations carry the analysis: depth-binned singular-value spectra of the effective update (**spectrum**, the pinned primary; 384 dimensions) and a canonicalized probe-projection (**probe**, corroborating; 12,672 dimensions). The decision rule was pinned by the Director before the bank completed: H2 (transfer fails) is supported iff, for both directions in the shift-controlled representation, (i) cross-family accuracy is not significantly above chance at one-sided exact-binomial $\alpha = 0.01$, and (ii) within-family accuracy exceeds cross-family accuracy by ≥ 15 percentage points.

The triviality control asks, before reading any transfer number, whether a classifier can tell *which family* an adapter came from. On raw representations it can, perfectly: family-identity probe 1.0000 (spectrum and probe, chance 0.5000). A classifier trained on one family and tested on the other is therefore tested under total covariate shift—raw “transfer-at-chance” would confirm our prediction while measuring nothing but family scale. Unsupervised per-family z-standardization (task labels never touched, so it has no capacity to manufacture task structure) collapses the probe to 0.1521 (spectrum) / 0.0000 (probe); both sit below binary chance, the signature of standardizing on the full pool before cross-validation—the family-*mean* separation the raw probe exploited is gone, and the transfer verdict rests on the binomial tests of the transfer accuracies themselves.

Table 2: Cross-family transfer accuracy, raw vs family-standardized. Chance = 0.1667. The headline is the standardized column, per the pinned rule; raw is descriptive. Margin = within-family minus cross-family accuracy (within-family accuracy is 1.0000 in every row).

representation	direction	raw	standardized	binom p (std)	margin
spectrum	qwen→llama	0.1667	0.7833	7.70×10^{-98}	21.67pp
spectrum	llama→qwen	0.1667	0.7375	1.20×10^{-84}	26.25pp
probe	qwen→llama	0.1750	0.7792	1.37×10^{-96}	22.08pp
probe	llama→qwen	0.2167	0.7792	1.37×10^{-96}	22.08pp

With the artifact removed, the picture inverts (Table 2). Raw transfer sits at or near chance in every direction—the number that would have “confirmed” the prediction. Standardized transfer runs 0.7375–0.7833 across a 1.5B and a 1B model of different lineages, with binomial $p \leq 1.20 \times 10^{-84}$; spectrum and probe agree. One caveat: the exact-binomial test treats the 240 test predictions per direction as independent, but they cluster by task, so the printed p -values are anti-conservative; the verdict does not lean on the exponents—accuracies of 0.74–0.78 against chance 0.1667 clear $\alpha = 0.01$ under any clustering-robust reading, and the ≥ 15 pp margin condition fails independently of the test. Because a pre-registration that flips via a control added in review carries the highest motivated-reasoning risk in the set, the Director re-ran the entire H2 pipeline end-to-end from the feature matrices, applying the standardization independently: every value reproduced to the digit. Had we run H2 without the control, we would now be reporting a confirmed prediction that is actually a measurement of family scale.

3 What it means

One mechanism, two results. Nuisance variation—the $GL(r)$ factorization gauge within a family, the scale gauge between families—dominates raw weight coordinates and obscures task structure that is intact underneath. Task identity is gauge-obscured, not absent. The scope of this claim runs on the label-granularity axis: our six coarse tasks are the opposite regime from W2T’s fine-grained attribute label spaces (312 binary attributes over 10k+-adapter collections) [3], and we make no claim about what a linear probe recovers, raw or canonical, at that granularity. Within our regime the statement is sharp: no learned encoder is needed—explicit gauge removal plus a linear SVM is already at ceiling, and the same features left in the raw gauge fail a $1.5\times$ -chance lock. This complements, rather than contradicts, the readout-at-scale line: weight-space readout of training provenance predates LoRA [1, 2], W2T [3] established LoRA checkpoints as legible objects under $QR \rightarrow SVD$ canonicalization at fine granularity, Learning on LoRAs [4] identified the $GL(r)$ symmetry and built equivariant processors for it, and spectral readouts [5] classify training objective from adapter spectra. Both ends of the granularity axis agree that the raw parameterization is the obstacle.

3.1 The identity backbone is the sole load-bearing structure (D2/H3)

D2 measures what the trained bridge is worth at inference time by transplanting bridges between adapters and measuring the val-loss consequence, under a locked directional hypothesis: the *substrate* hypothesis (swaps cost little) against the *task-specific* hypothesis (cross-task penalties are large and scale with task distance). $K = 3$ donor/recipient pairs per cell give 360 evaluations per family (18 native references; penalty cells $n = 90$ for cross-task/magnitude/topology, $n = 18$ elsewhere), every one re-assembled from the bank and SHA-verified before evaluation; all passed. Seven kinds are evaluated, of which two carry the structural contrast: **permuted_deviation** scrambles the trained deviation while preserving the identity backbone; the full-entry permutation (a shared derangement of all 36 bridge-entry positions) destroys the backbone. The two share the same derangement per cell, so their difference isolates the backbone effect and nothing else.

Table 3: D2 penalty matrix (mean val-loss penalty vs native, nats). The **permuted** means carry per-eval spread: SD 2.31 (qwen) / 3.34 (llama) at $n = 18$.

swap kind	qwen2.5-1.5b	llama3.2-1b
cross_seed	+0.0000	+0.0002
cross_task	+0.0000	+0.0003
cross_task_magnitude	+0.0000	+0.0000
cross_task_topology	+0.0000	+0.0002
identity	+0.0000	+0.0007
permuted_deviation (H3 reference)	+0.0000	+0.0007
permuted (full)	+2.8086	+3.8365

Every swap that preserves the identity backbone costs less than +0.001 nats—installing a different task’s bridge, deleting the trained bridge entirely, or scrambling the trained deviation. Only the full-entry permutation costs anything: +2.8086 / +3.8365 nats (Table 3). The substrate hypothesis is supported and the task-specific hypothesis refuted; the scope is in-distribution loss under the house bridge architecture. The Director reproduced all 14 per-kind means exactly from the per-eval rows.

3.2 Post-hoc, weight-only merge-conflict prediction (D3)

D3 asks whether two adapters’ weights alone—no training access, no activations, no data—predict whether their merge degrades. The regime matters: the training-time form of merge-conflict prediction is owned by [6]; D3 claims only the weight-only, post-hoc regime, and weight-only conflict *heuristics* (sign agreement, task-vector geometry) predate it [7, 8]. The quantity predicted—whether the $\alpha = 0.5$ midpoint interpolation degrades endpoint loss—is the loss barrier of the linear-mode-connectivity literature [9, 10].

Design: $N = 120$ vertex-disjoint pairs per family (each run used at most once—the dyadic-dependence-safe design), $\alpha = 0.5$ midpoint merges, 240 merged adapters GPU-evaluated on both endpoints. Label

rule, pinned before the bank completed: conflict iff the merge degrades either endpoint by $\geq 5\%$ in relative perplexity, with a 10% degenerate-balance floor (fallback not used). The realized conflict rate is itself a finding about midpoint merging in this bank: pooled 85.8%; qwen 99 conflicts / 21 non-conflicts (82.5%), llama 107 / 13 (89.2%), llama’s minority fraction (10.8%) clearing the floor by a single pair. Features: a 2-scalar **distance** baseline (cosine and ℓ_2) against a **full** weight-only set adding 4 gauge-invariant principal-angle aggregates and two per-module vectors (length 112 Qwen / 64 Llama). Classifier: logistic regression, group-aware StratifiedGroupKFold, fold seed 0, 5 splits, 1000 bootstrap replicates—all pinned.

Table 4: D3 merge-conflict prediction, group-aware CV AUC with 95% bootstrap CIs (120 pairs/family; pooled OOF row descriptive).

family	AUC full (weight-only)	AUC distance-only	full – distance
qwen2.5-1.5b	0.995 [0.983, 1.000]	0.675 [0.484, 0.848]	+0.320 [0.150, 0.511]
llama3.2-1b	0.962 [0.898, 0.999]	0.713 [0.458, 0.923]	+0.249 [0.039, 0.490]
pooled OOF (descriptive)	0.9890 [0.9730, 0.9992]	0.7082	+0.2808 [0.1565, 0.4205]

Two adapters’ weights alone predict whether their midpoint merge degrades an endpoint at AUC 0.96–0.99, and the gauge-invariant block adds a margin over raw distance whose CI excludes zero in both families (Table 4). Two caveats bind the headline. First, the distance-only baseline is fold-scheme-sensitive: the Director’s independent from-scratch re-run under the naive scheme reproduced the naive block exactly (distance 0.686 / 0.667, full 0.995 / 0.952) against the group-aware headline baseline of 0.675 / 0.713—a ~ 0.01 – 0.05 fold-scheme gap on a 2-feature model, which is why the fold configuration is pinned and the fragility reported. Second, the conflict label is strongly associated with task composition (negatives are dominated by same-task pairs: 13 of 21 qwen, 9 of 13 llama), and H1 shows task identity is fully recoverable from weight features; a task-composition oracle alone reaches AUC 0.804 / 0.785—above the distance baseline—so the headline AUC upper-bounds the predictor’s discriminative value within the cross-task-only regime. The full model exceeds the oracle bound, so geometric signal beyond task identity is present, but we bound the within-regime value rather than claim it.

3.3 The pilot correlation shrinks honestly (D-aux)

D-aux re-verifies, at bank scale, a pilot correlation of $r = 0.888$ between bridge deviation (**dev_mean**) and generalization gap (**final_gap**)—the re-verification itself pinned in the Director decisions before the bank completed. A step-0 identity control must read exactly 0.0 by construction (identity-initialized bridge); it did.

Table 5: D-aux bridge-deviation $\leftrightarrow$ generalization-gap correlation (Pearson r , 95% bootstrap CIs). Within-task cells are the pre-registered Simpson’s-paradox guard.

cell	n	Pearson r	95% CI
pooled	480	0.300	[0.175, 0.415]
qwen2.5-1.5b	240	0.418	[0.323, 0.522]
llama3.2-1b	240	0.337	[0.201, 0.466]
within math (llama)	40	0.549	[0.324, 0.736]
within math (qwen)	40	0.336	[0.030, 0.652]
within xsum (qwen)	40	−0.301	[−0.502, −0.066]

The pilot’s $r = 0.888$ does not survive the bank, and we report that as the result (Table 5): pooled over 480 runs the association is real, positive, and modest ($r = 0.300$ [0.175, 0.415]), inflated in the pilot by small- n and task mixture, and heterogeneous within task—math positive in both families, xsum in qwen significantly negative. A deviation measure whose within-task sign flips between cells is not a universal overfitting signal. Read with D2, the conjunction is the finding: the trained bridge deviation carries information *about* the run (enough for a weak correlate of its generalization gap) without carrying the in-distribution task solution, which lives in the identity backbone. The bridge is a diagnostic-bearing structure, not a load-bearing one.

4 Methods, compressed: pre-registration with teeth

Three layers, each with a paper trail. (1) *The locked card*: the hypotheses (H1, H2, H3, the D3 prediction target, the D-aux re-verification) were locked in the experiment card of 2026-07-03, before the bank existed. (2) *Director-pinned decisions*: every analytical choice the card left open was pinned in a dated ruling by an independent Director instance on 2026-07-06, before the bank completed—the H2 decision rule, the D2 structure reference (with the magnitude/topology decomposition cells run unconditionally, so that no data-dependent forking point exists), and the D3 label rule; the vocab-signature D1 arm entered as a dated amendment. The one post-completion declaration, the D3 pair design, was filed 2026-07-20 before any pairs, merges, or labels existed, amended the same day (still before any pair artifact) from per-cell stratification to the uniform vertex-disjoint sampler the frozen tool actually implements, and ratified by the Director as a clean dated amendment; the realized coverage of the 21 task-pair cells is reported descriptively (same-task pairs $14/120 = 11.7\%$ qwen, $22/120 = 18.3\%$ llama, against $\sim 16\% \approx 20/120$ expected under uniformity). (3) *The completeness interlock*: every CLI that can touch the real bank calls `require_complete_bank()` before reading a single adapter, passing only at 480/480 COMPLETE in the manifest. The tooling passed 171/171 tests plus the D1/D3/D-aux selftests on 2026-07-20, and the interlock HELD: no real-bank statistic was computed before 480/480. Each analysis then ran against the real bank exactly once.

Verification is maker-grader separation at three levels: fresh-context adversarial review of every custom component before it touched data (the D3 labels runner, written post-card, was verified before its GPU run, catching two blocking defects; its native-loss shortcut was accepted only after verifying 0.00000% divergence against 36 fresh D2-harness evaluations); the interlock as verifier of the analysts; and an independent Director regrade of every headline from per-item data. After release, the paper’s claims were dispositioned against a 117-finding adversarial audit (1 blocker, 30 major, 53 minor, 33 note, over 7 rounds and 19 lenses), with every finding resolved. Two of the five pre-registered outcomes—the H2 refutation and the D-aux shrink—went against the authors, and they are the credibility of the other three: a pipeline that can only confirm is not a measurement instrument. The operational discipline itself (typed state over prose restatement, maker-grader separation) is measured in [11]; we cite it for method provenance, not numbers.

5 Limitations

Cost. The bank consumed roughly 17 days of a single local GPU (an NVIDIA RTX 6000 Ada, 48 GB); measured per-run rates are 30.56 min/run (llama3.2-1b, $n = 240$), 51.72 min/run (qwen2.5-1.5b, $n = 240$), blended 41.14 min/run over all 480 [measured]. The analyses are cheap—D1, D-aux, and D3 feature extraction are CPU-only—but the artifact they require is not.

Scope. Two families at 1–1.5B scale; whether the ceilings hold at 7B+ or across architecture classes is unmeasured. Six coarse tasks: the label-granularity axis cuts both ways—we claim nothing about fine-grained regimes, and ceiling accuracy on six classes means H1 cannot rank representations above the lock. In-distribution val loss only: D2’s verdict is explicitly about this loss; behavioral and out-of-distribution consequences of swaps and merges are unmeasured. Single merge operator: $\alpha = 0.5$ midpoints only; other α , TIES/DARE-style operators, and >2 -adapter merges are outside the measured claim. One bridge architecture and one training recipe: D2 and D-aux are properties of the house identity-initialized, controller-free bridge, and every adapter shares a single hyperparameter configuration; heterogeneous banks could couple recipe to task and change every result.

Statistical fragilities, reported where they appear. The D3 distance baseline moves ~ 0.01 – 0.05 with fold scheme; the fold configuration is pinned. The D3 negative class is small (21 qwen / 13 llama non-conflict pairs; llama clears the 10% floor by a single pair), and the task-composition oracle (AUC 0.804 / 0.785) bounds how much of the headline is task identification. The D-aux pooled $r = 0.300$ conceals sign-varying within-task structure ($n = 40$ per cell); any within-task use would need task-specific calibration the bank is only pilot-scale for.

6 Reproducibility

Every headline number is re-derivable from a per-item verification bundle, and every one has already been re-derived by someone who was not us. The bundle carries per-item result data (confusion matrices and permutation summaries behind every H1 accuracy; the full H2 transfer and family-identity tables; 360 per-eval D2 rows per family with assembly SHA-256s; 240 per-pair D3 feature dicts and labels; 480 per-run D-aux pairs) and the reduced feature matrices (`features_<family>.npz`, 90.6 MB qwen / 56.7 MB llama, float32), exported by the frozen feature functions so a from-scratch classifier re-run bit-matches. The bundle is anchored at rhombic commit `638f4a8`, archive SHA-256 `c1891d50fc7c48c27d6fb606d667f1f1120105ed55cb7414c8e4b3d874acfadd`, all 16 files matching the SHA256SUMS manifest byte-for-byte. The verification bundle and all analysis code ship in the public repository (<https://github.com/tasumermaf/rhombic>); the feature matrices (~147 MB total) are published as a Hugging Face dataset linked from the repository README. The adapter bank itself remains a local artifact for size; its manifest, per-run configs, and the trainer that reproduces it are in the repository. The parent paper (`paper/rhombic-asset1.tex` in the same repository) carries the full design, decision record, and audit trail this condensation compresses.

Acknowledgments

Experimentation, analysis, and drafting were performed with a standing AI research collaborator (Claude, Anthropic); the Director role was performed by a separate instance with no shared context, whose dated rulings and regrades are archived in the repository. All numbers reported here were machine-verified against the frozen artifacts rather than restated from any model’s summary.

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